

Homework 4

Due Friday February 27, 2026

Reading:

- Required: Lecture 6 Notes
- Optional: PGMCT Ch. 6 and GMEFVI Ch. 6

Problems:

1. (LEARNING WITH A SINGLE VARIABLE AND FACTOR). Let $X \in [m] := \{1, 2, \dots, m\}$ and consider the one-factor model

$$p^\theta(x) = \frac{1}{Z(\theta)} \exp(\theta_x), \quad Z(\theta) = \sum_{u=1}^m \exp(\theta_u),$$

where $\theta = (\theta_1, \dots, \theta_m) \in \mathbb{R}^m$. Given i.i.d. data $\mathcal{D} = \{x^{(1)}, \dots, x^{(N)}\}$, define counts

$$N_x := \sum_{\ell=1}^N \mathbf{1}\{x^{(\ell)} = x\}, \quad \hat{p}_{\mathcal{D}}(x) := \frac{N_x}{N}.$$

- (a) Write the normalized log-likelihood $\frac{1}{N} \mathcal{L}(\theta)$ explicitly in terms of $\{N_x\}$ and θ .
 - (b) Compute $\frac{\partial}{\partial \theta_x} \frac{1}{N} \mathcal{L}(\theta)$ and simplify it to the difference between data and model terms.
 - (c) Show that $\frac{1}{N} \mathcal{L}(\theta)$ has a maximizer that is not unique due to an extra degree of freedom.
 - (d) Impose the constraint $\sum_{x=1}^m \theta_x = 0$. Under this constraint, solve for the MLE $\hat{\theta}$ in closed form in terms of $\hat{p}_{\mathcal{D}}$.
2. (IDENTIFIABILITY AND GAUGE). Consider a factor graph with indicator/table factors

$$f_a(x_{\partial a}; \theta_a) = \exp(\theta_{a, x_{\partial a}}), \quad \theta_a \in \mathbb{R}^{|\mathcal{X}_{\partial a}|}, \quad \theta_{a, x_{\partial a}} = [\theta_a]_{x_{\partial a}}.$$

Assume the graph is connected and all variables are discrete.

- (a) Show that if you add a constant c_a to *each* entry of θ_a (possibly different across factors), the distribution $p^\theta(x)$ does not change. Thus, the constants c_a cannot be identified from data.
- (b) Show that there is at least one additional redundancy: a “reparameterization” that preserves p^θ in a pairwise MRF. Specifically, consider a pairwise MRF

$$p^\theta(x) \propto \exp\left(\sum_{i \in V} \theta_i(x_i) + \sum_{(i,j) \in E} \theta_{ij}(x_i, x_j)\right).$$

Show that we can use arbitrary functions $\phi_i : \mathcal{X}_i \rightarrow \mathbb{R}$ to define new parameters θ'_i, θ'_{ij} (depending on ϕ_i) so that

$$\sum_i \theta'_i(x_i) + \sum_{(i,j)} \theta'_{ij}(x_i, x_j) = \sum_i \theta_i(x_i) + \sum_{(i,j)} \theta_{ij}(x_i, x_j) \quad \text{for all } x.$$

- (c) Propose a set of constraints that removes these redundancies (not necessarily uniquely), and briefly argue why it fixes identifiability.
3. (EM SOFT CLUSTERING). Let $Y \in \{1, \dots, m\}$ be observed and $H \in \{1, \dots, K\}$ hidden. Consider

$$p^\theta(y, h) \propto \exp(\theta_{yh} + \alpha_y + \beta_h).$$

One can interpret H as a latent class (or cluster) index and $q^{(\ell)}(h)$ as a soft cluster assignment for datum $y^{(\ell)}$.

- (a) Show that $p^\theta(h | y)$ is a softmax in the scores $\theta_{y*} + \beta$.
 - (b) For data $\{y^{(\ell)}\}_{\ell=1}^N$, write the E-step responsibilities and explain their clustering meaning.
 - (c) Derive the M-step objective and first-order condition in terms of soft counts.
 - (d) Describe explicit constraints for (θ, α, β) that remove symmetries to make the parameters more identifiable.
4. (SMALL-STEP VIEWPOINT: EM DIRECTION AS LIKELIHOOD GRADIENT.) Let $\mathcal{L}_{\text{obs}}(\theta) = \sum_{\ell=1}^N \ln p^\theta(y^{(\ell)})$ be the observed-data log-likelihood in a latent-variable model with latent H and complete-data log-density $\ln p^\theta(y, h)$.

Fix a current parameter θ and define the *Q-function*

$$Q(\theta' | \theta) := \sum_{\ell=1}^N \mathbb{E}_{p^\theta(H|y^{(\ell)})} [\ln p^{\theta'}(y^{(\ell)}, H)].$$

- (a) Show that the gradient of the observed-data log-likelihood at θ equals the gradient of $Q(\theta' | \theta)$ with respect to θ' evaluated at $\theta' = \theta$:

$$\nabla_\theta \mathcal{L}_{\text{obs}}(\theta) = \nabla_{\theta'} Q(\theta' | \theta) \big|_{\theta'=\theta}.$$

(Assume you can interchange derivative and summation/expectation.)

- (b) Conclude that if you do a *small* M-step update of the form

$$\theta^{\text{new}} = \theta + \eta \nabla_{\theta'} Q(\theta' | \theta) \big|_{\theta'=\theta} \quad \text{with } \eta > 0 \text{ small,}$$

then θ^{new} is a first-order ascent step for $\mathcal{L}_{\text{obs}}(\theta)$.

- (c) (Interpretation) Explain in one paragraph why this supports the intuition that “the E-step is exact in the small-change limit.”
5. (A LINEAR-PROGRAMMING SOLVER FOR SUDOKU). Combinatorial problems that optimize over permutations (e.g., scheduling and assignment problems) can be quite challenging computationally. In this problem, we explore low-complexity linear-programming (LP) relaxations.

- (a) Assume n tasks must be assigned to n agents and let $x_{i,j}$ denote the indicator variable for “task i assigned to agent j ”. If the cost of assigning task i to agent j is $c_{i,j}$, then the best assignment problem is equivalent to the integer linear program

$$\min_{\{x_{i,j}\}} \sum_{i,j} c_{i,j} x_{i,j} \quad \text{subject to} \quad x_{i,j} \in \{0,1\}, \sum_i x_{i,j} = 1, \sum_j x_{i,j} = 1.$$

Solve this problem by hand for $n = 4$ and

$$C = \begin{bmatrix} 10 & 7 & 24 & 6 \\ 19 & 8 & 7 & 8 \\ 16 & 22 & 12 & 15 \\ 4 & 8 & 10 & 14 \end{bmatrix}.$$

Now, solve the linear program where the constraint $x_{i,j} \in \{0,1\}$ is relaxed to $x_{i,j} \in [0,1]$. This minimizes average cost when $x_{i,j}$ is the probability of assigning task i to agent j .

- (b) Using a similar approach, an $N \times N$ Sudoku game (where N is a perfect square) can be cast an integer linear program with N^3 variables. Let $x_{i,j,k}$ denote the indicator variable for “the square in row i and column j is assigned the value k ”. Based on the previous part, write the permutation constraints as linear equalities for $N = 4$.
- (c) The Sudoku puzzle is solved by computing

$$\min_{\{x_{i,j,k}\}} \sum_{i,j,k} c_{i,j,k} x_{i,j,k},$$

where $c_{i,j,k}$ be the cost of assigning the integer k to the (i,j) -th entry of the puzzle. Known values in the puzzle can be handled either adding equality constraints or by adjusting the cost function. For example, “the square in row i and column j is observed to be k ” implies the equality constraint $x_{i,j,k} = 1$. Known values can also be included by initializing $c_{i,j,k}$ to all zeros and then setting $c_{i,j,k} = -1$ for each (i,j,k) observation. Choose one of these approaches and use linear programming to solve the $N = 4$ Sudoku:

	2		
1		2	
2		3	
	3		

- (d) Let $\mathcal{X} = \{1, 2, \dots, N\}$ and $f : \mathcal{X}^N \rightarrow \{0,1\}$ be a factor defined by $f(z_1^N) = 1$ if z_1^N defines a permutation of $\mathcal{X}$ and 0 otherwise. From this, one can construct the Sudoku factor graph by introducing N^2 variables and $3N^2$ permutation factor nodes. Local 0/1 factors can be used to account for known values. In this case, marginalization gives

$$g_i(x_i) = \sum_{\underline{x} \setminus x_i} f(\underline{x}) = \# \text{ valid patterns matching all observations with fixed } x_i.$$

Unfortunately, naive implementation of the sum-product algorithm for the permutation factor requires the summation of $N!$ terms. Describe how one might implement a factor-graph permutation constraint with much lower complexity. [Hint: It is fine to introduce additional variable and factor nodes.]

Practice Problems (do not hand in):

1. (BIT MAP VERSUS BLOCK MAP DECODING). Consider the binary linear code $\mathcal{C}$ defined by the parity-check matrix

$$H = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}.$$

- (a) List the four codewords in $\mathcal{C}$.
- (b) A codeword $\underline{X} = (X_1, \dots, X_5) \in \mathcal{C}$ is chosen uniformly at random from $\mathcal{C}$ and transmitted over a binary symmetric channel with cross-over probability $p = 1/3$. Using the sum-product algorithm, determine $\Pr(X_3 = 0 | \underline{Y} = (1, 1, 1, 1, 1))$, i.e., the probability that $X_3 = 0$ given that the particular vector $\underline{Y} = (1, 1, 1, 1, 1)$ is received. Clearly show the values of messages passed on the edges of the factor graph during each step of the sum-product algorithm.
- (c) Let $\mathcal{C}_0 = \{\underline{x} \in \mathcal{C} \mid x_3 = 0\}$ and $\mathcal{C}_1 = \{\underline{x} \in \mathcal{C} \mid x_3 = 1\}$. Show that

$$\frac{\Pr(X_3 = 0 | \underline{Y} = \underline{y})}{\Pr(X_3 = 1 | \underline{Y} = \underline{y})} = \frac{\sum_{\underline{x} \in \mathcal{C}_0} \Pr(\underline{Y} = \underline{y} | \underline{X} = \underline{x})}{\sum_{\underline{x} \in \mathcal{C}_1} \Pr(\underline{Y} = \underline{y} | \underline{X} = \underline{x})}$$

and use this expression to verify your answer to (b).

- (d) Suppose that channel crossover probability in the previous question is changed to $p = 2/5$. Show that the symbol-wise MAP decision for X_3 does not agree with the block-wise MAP decision for $\underline{X}$.
2. (MESSAGE PASSING AS REPARAMETERIZATION). Consider a strictly positive pairwise MRF

$$p(x) \propto \prod_{i \in V} \psi_i(x_i) \prod_{(i,j) \in E} \psi_{ij}(x_i, x_j).$$

For each directed edge $i \rightarrow j$, let $\rho_{i \rightarrow j} : \mathcal{X}_j \rightarrow \mathbb{R}_{>0}$ be any positive function (note: the argument is a value z_j of the destination variable). Define reparameterized factors

$$\psi'_i(x_i) := \psi_i(x_i) \prod_{k \in \partial i} \rho_{k \rightarrow i}(x_i), \quad \psi'_{ij}(x_i, x_j) := \psi_{ij}(x_i, x_j) \frac{1}{\rho_{i \rightarrow j}(x_j) \rho_{j \rightarrow i}(x_i)}.$$

- (a) Show that this transformation preserves the joint distribution exactly.
- (b) Explain why this is a gauge/reparameterization symmetry and how BP fits into it.

(c) At a BP fixed point, define beliefs

$$b_i(x_i) \propto \psi_i(x_i) \prod_{k \in \partial i} m_{k \rightarrow i}(x_i),$$

$$b_{ij}(x_i, x_j) \propto \psi_{ij}(x_i, x_j) \psi_i(x_i) \psi_j(x_j) \prod_{k \in \partial i \setminus \{j\}} m_{k \rightarrow i}(x_i) \prod_{\ell \in \partial j \setminus \{i\}} m_{\ell \rightarrow j}(x_j).$$

Show briefly how fixed-point equations imply local calibration

$$\sum_{x_i} b_{ij}(x_i, x_j) = b_j(x_j), \quad \sum_{x_j} b_{ij}(x_i, x_j) = b_i(x_i).$$

(d) Explain what is special about the BP fixed-point parameterization.

3. (CONSISTENCY OF MAXIMUM PSEUDO-LIKELIHOOD (MPL) ESTIMATION). Let $X_i \in [q]$ and consider a pairwise FG:

$$p^\theta(x) \propto \exp\left(\sum_{i \in V} \theta_i(x_i) + \sum_{(i,j) \in E} \theta_{ij}(x_i, x_j)\right).$$

Define the pseudo-log-likelihood to be

$$\ell_{\text{PL}}(\theta) = \frac{1}{N} \sum_{\ell=1}^N \sum_{i \in V} \ln p^\theta(x_i^{(\ell)} | x_{\partial i}^{(\ell)}).$$

Assume data are i.i.d. from a true parameter θ^* (up to gauge) and all conditionals are strictly positive.

- (a) Derive $p^\theta(x_i | x_{\partial i})$ explicitly in terms of θ_i and θ_{ij} .
- (b) Compute $\frac{\partial}{\partial \theta_i(a)} \ell_{\text{PL}}(\theta)$.
- (c) Compute $\frac{\partial}{\partial \theta_{ij}(a,b)} \ell_{\text{PL}}(\theta)$ for an edge $(i, j) \in E$.
- (d) Let $p^\mathcal{D}$ be the empirical distribution of the data and define the *population* pseudo-likelihood

$$\mathcal{L}_{\text{PL}}^{\text{pop}}(\theta) := \sum_{i \in V} \mathbb{E}_{p^\mathcal{D}}[\ln p^\theta(X_i | X_{\partial i})].$$

Now, assume that $p^\mathcal{D} \approx p^{\theta_0}$ and the above to derive

$$\mathcal{L}_{\text{PL}}^{\text{pop}}(\theta_0) - \mathcal{L}_{\text{PL}}^{\text{pop}}(\theta) = \sum_{i \in V} \mathbb{E}_{X_{\partial i} \sim p^{\theta_0}} \left[D\left(p^{\theta_0}(X_i | X_{\partial i}) \| p^\theta(X_i | X_{\partial i})\right) \right] \geq 0.$$

Explain what this implies about MPL estimation.

4. (COMPLEXITY OF LEARNING WITH BP INNER LOOP.) Consider a tree-structured factor graph with maximum variable alphabet size q and maximum factor degree d .

- (a) Give the time complexity (big- O) of one full BP pass (all directed messages) in terms of n, q, d .
 - (b) Now consider a single cycle of length n with pairwise factors and alphabet size q . If loopy BP runs for T iterations, give the complexity.
 - (c) Explain briefly why learning by gradient ascent on loopy graphs can be far more expensive than on trees even if each BP iteration is cheap.
5. (LEARNING WITH A SINGLE CYCLE). Consider an inhomogeneous binary Ising model ($x_t \in \{\pm 1\}$) on an n -cycle:

$$p^{J,h}(x) \propto \exp\left(\sum_{t=1}^n J_t x_t x_{t+1} + \sum_{t=1}^n h_t x_t\right), \quad x_{n+1} \equiv x_1.$$

Here each edge has its own coupling J_t and each node its own field h_t . Assume N i.i.d. samples $\{x^{(\ell)}\}_{\ell=1}^N$.

- (a) Write the normalized log-likelihood $\frac{1}{N}\mathcal{L}(J, h)$ and its gradient in componentwise “data moment minus model moment” form.
- (b) Show how to compute all model moments

$$\mathbb{E}_{p^{J,h}}[X_t], \quad \mathbb{E}_{p^{J,h}}[X_t X_{t+1}], \quad t = 1, \dots, n,$$

exactly in $O(n)$ time using “break the cycle by conditioning on one spin.”

- (c) Give a concrete gradient-ascent algorithm using this exact inference routine.
- (d) (Insight) Why is one-cycle learning much easier than 2D-grid learning, despite both being loopy?